VEDA Technology Internship — Task 21 Report

Responsive Navbar with Hamburger Menu Documentation & Verification (Light Pastel Green & White Theme)

1. Project Overview

This report documents the design, architecture, and verification of **Task 21 (Responsive Navbar with Hamburger Menu)** for the VEDA Technology Web Development Internship. The project delivers a fully responsive, accessible, and high-performance navigation component engineered strictly with **HTML5**, **CSS3**, and **Vanilla JavaScript (ES6+)**. Styled with an elegant **Light Pastel Green and White** theme (#F4F8F5 background, #FFFFFF navbar/cards, #2D6A4F primary forest green, #EAF5EE badge accents, #D0E7D7 borders), it provides a desktop horizontal navigation bar and collapses into an animated 3-bar hamburger drawer on mobile screens (≤ 768px). The menu features smooth slide-in/out transitions, automatic closure on link selection or backdrop clicks, Escape key dismissal, and comprehensive WAI-ARIA states with zero external dependencies.

2. Technical Architecture & Component Stack

Component	Implementation & Technical Details
HTML5 Semantic Structure	Structured with <header>, <nav>, , , <a>, and <button> using proper accessibility landmarks and ARIA attributes.
Pastel Green & White Theme	Light theme (#F4F8F5 / #FFFFFF) with sage green accents (#2D6A4F, #EAF5EE), high-contrast accessible text (#1C2B21), visible focus rings, and clean mobile-first layout.
Animated Hamburger Button	3 horizontal span lines smoothly morphing into an "X" via CSS transforms (rotate 45deg, scaleX 0, translateY) when .is-active is applied.
Hardware Slide Transitions	Off-canvas drawer utilizing CSS transform: translateX(105%) to translateX(0) with 0.28s cubic-bezier timing for 60fps sliding.
classList.toggle() Control	Menu state toggling strictly powered by vanilla JS classList.toggle('is-active') and classList.toggle('is-open').
Auto-Close & Escape Key	Drawer collapses automatically upon clicking any nav link, clicking the backdrop overlay, or pressing the Esc key.
WAI-ARIA & Focus Recovery	Dynamic updates for aria-expanded and aria-label; focus automatically returns to the toggle button on Escape.
Zero Dependencies	100% pure vanilla HTML5, CSS3, and JavaScript; zero frameworks, zero external dependencies, ready for instant GitHub Pages hosting.

3. Functional Specifications Breakdown

Requirement	Logic & Implementation Description	Status
1. Desktop Horizontal Bar	Displays brand and links in a clean horizontal row on screens > 768px with pill hover effects.	**PASSED**
2. Brand & 5 Nav Links	Includes VEDATech brand with logo and links: Home, About, Services, Projects, Contact.	**PASSED**
3. Mobile Hamburger Menu	Collapses links into an off-canvas drawer on mobile screens (≤ 768px).	**PASSED**
4. Hamburger Click Toggle	Clicking hamburger button opens and closes mobile menu using classList.toggle().	**PASSED**
5. CSS Media Queries	Responsive layout switching at 768px and 480px breakpoints using pure CSS media queries.	**PASSED**
6. Slide Transitions	Smooth 0.28s cubic-bezier slide-in and slide-out transition without layout shift or jank.	**PASSED**
7. Auto-Close on Link Select	Clicking any navigation link closes mobile menu immediately and scrolls to section.	**PASSED**
8. Escape Key Support	Pressing Escape key closes drawer and returns focus to the toggle button.	**PASSED**

9. Animated Hamburger Icon	Updates hamburger icon: 3 lines morph into an "X" when open, reverting on close.	**PASSED**
10. Accessible ARIA & Focus	aria-expanded, aria-label, aria-controls, and high-contrast visible keyboard focus rings.	**PASSED**

4. Verification & Testing Matrix

Test Scenario	Input Action	Expected State	Actual State	Status
Desktop Viewport	Window width > 768px	Horizontal navbar visible, hamburger button hidden	Links displayed horizontally, toggle hidden	**PASSED**
Mobile Viewport	Window width ≤ 768px	Links collapsed into drawer, hamburger button visible	Drawer hidden off-screen, hamburger visible	**PASSED**
Open Hamburger	Click hamburger button	Drawer slides in, hamburger morphs to "X", overlay dims	Drawer visible, "X" icon active, overlay active	**PASSED**
Close Hamburger	Click "X" button	Drawer slides out, icon returns to 3 bars, overlay hidden	Drawer closed, 3 bars restored, overlay cleared	**PASSED**
Link Selection	Click "Services" link	Menu closes automatically, page scrolls to #services	Drawer closed, smoothly scrolled to section	**PASSED**
Escape Key Press	Press Escape key	Drawer closes instantly, focus returned to toggle	Menu closed, focus restored to button	**PASSED**
Backdrop Click	Click outside drawer	Drawer dismisses smoothly, overlay fades out	Menu closed, overlay dismissed cleanly	**PASSED**
Window Resize	Resize mobile to desktop	Open drawer classes cleared, desktop layout restored	Clean desktop reset without stuck classes	**PASSED**

5. Deployment & Repository Structure

• **Directory:** Task21-Responsive-Navbar/

• **Files:** index.html, style.css, script.js, README.md

• **GitHub Repository:** https://github.com/dharmarayank946/Responsive-Navbar-with-Hamburger-Menu.git

• **GitHub Pages Live Demo:** https://dharmarayank946.github.io/Responsive-Navbar-with-Hamburger-Menu/

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